Project Initialization and Planning Phase

Date	24 June 2026
Team ID	
Project Title	EduGenie - AI-Powered Educational Learning Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report outlines the design and implementation of EduGenie, an AI-powered educational learning assistant. It leverages advanced language models (Google Gemini 1.5 Pro and LaMini-Flan-T5) to provide personalized learning support, interactive quiz generation, notes summarization, and RAG-based document chat to enhance student comprehension and track progress.

Project Overview	
Objective	Provide students, teachers, and self-learners with an intelligent, personalized study assistant that leverages advanced generative AI to answer questions, explain concepts, generate quizzes, and suggest custom learning paths.
Scope	Full-stack educational platform with AI interfaces for Q&A, multi-level definitions, automated note summaries, PDF/DOCX file uploads for contextual reading, and progress gamification (streaks, badges).
Problem Statement	
Description	Traditional learning materials lack personalization, making it difficult for students of varying skill levels to grasp complex topics, find high-quality practice quizzes, or map out structured learning plans.
Impact	Improves student understanding, provides targeted practice assessments, saves time on summarizing notes, and increases study motivation through progress tracking and custom learning roadmaps.
Proposed Solution	
Approach	Building a FastAPI backend with Jinja2 frontend, integrating the Gemini API for generative tasks and a local or API fallback LaMini model for multi-level topic explanations, with structured JSON storage.
Key Features	AI Question Answering (with PDF/DOCX RAG support), Multi-level Concept Explainer, Interactive Quiz Generator (with scoring and PDF export), Document Summarizer (different summary styles), Personalized Learning Path Generator (roadmaps with weeks, milestones, projects), Progress Analytics (streaks, study streak tracker).

	User profile configuration with custom settings (learning level, preferred explanation format) and historical bookmarking/history search.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	JSON database storage files (users, queries, summaries, quizzes, roadmaps, analytics)